



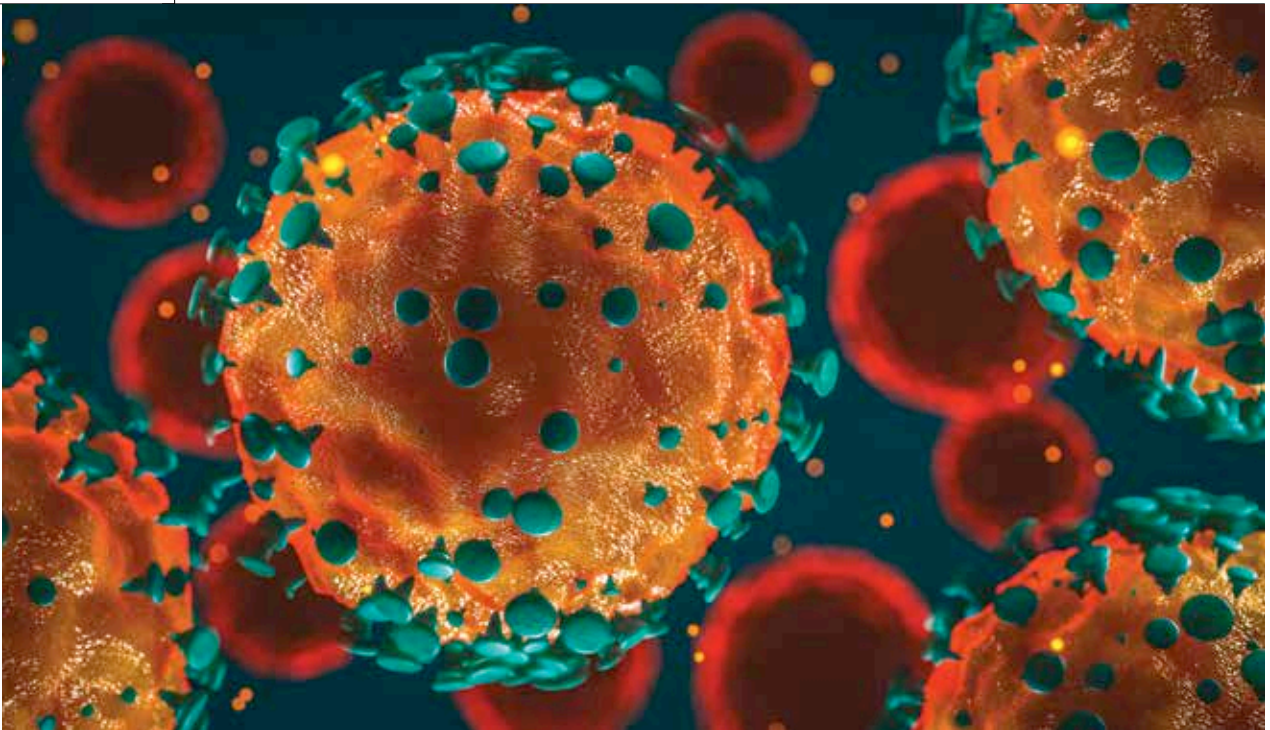
App analyses coronavirus

An app named Genopo can be used to analyse the genome of the SARS-CoV-2 virus on a phone. <http://digit.in/oct20-35>



Social distancing > travel ban

As per a study, social distancing is more effective at keeping the pandemic at bay than travel bans. <http://digit.in/oct20-36>



FIGHTING THE PANDEMIC

PART II

The research directions to stop the spread of the novel coronavirus

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t the time of writing, there were 212 vaccines being developed for the novel coronavirus,

with around 9 vaccine candidates undergoing phase III clinical trials. Scientists around the world have redirected their efforts towards tackling the pandemic, that seems to show no signs of letting up. Labs

around the world are developing faster and more accurate tests, understanding how the behaviour of people is changing because of the pandemic, exploring new treatment procedures, better understanding the long term effects of COVID-19, as well as investigating the exact mechanisms of how the SARS-CoV-2 virus works.

TREATMENTS

A team of international researchers at the University of Buffalo are investigating the use of an inexpensive and widely available drug, known as melatonin, to treat milder forms of the disease. Melatonin's anti-inflammatory, anti-oxidation, and immune boosting properties are suitable for tackling COVID-19. Melatonin promises to suppress the cytokine shock, an aggressive immune system response to the virus that causes fluid build up in the lungs, and leads to extreme cases. Most of

the research and treatments that have emerged in the past few months are focused on tackling the more aggressive manifestations of the disease. The new clinical trial and proposed treatment would allow better care for those not affected severely enough to be hospitalised, and also help slow down the spread of the disease.

Researchers use something called adjuvants to increase the efficacy of vaccines, especially for harder to treat diseases such as dengue or HIV. Adjuvants are pharmacological agents that help boost the immune system reaction. Adjuvants however, can cause swelling at the point where they are injected, and at the same time trigger the more extreme forms of the immune system response, which is one of the things that makes COVID-19 so dangerous. Researches from the University of Chicago have found a molecule, a peptide, that when used in tandem with a vaccine, disrupts pathways in cells that causes the inflammation at the site of injection, as well as an extreme immune response called the cytokine storm. The efficacy of an existing flu vaccine was boosted with the introduction of the molecule. The researchers are now looking for